SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1710**

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

Time Duration: 1 Hour
Total Marks:30

QUESTIONS: 5

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

192565026

I Shaik Fahed *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Annexure A

Resolution Algorithm Implementation

Student 31: Shaik Fahed | Register Number: 192565026

Question 1 - Account Verification

Knowledge Base

1. If a user uploads a valid ID, then the account is verified.
2. The user uploaded a valid ID.

Goal

Prove that **the account is verified.** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 2 - Premium Subscription Access

Knowledge Base

1. If a customer pays the subscription fee, then the customer gets premium access.
2. Anitha paid the subscription fee.

Goal

Prove that **Anitha gets premium access.** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 3 - Smart Device Connectivity

Knowledge Base

1. If a device is connected to WiFi, then the device can access the internet.
2. The smart TV is connected to WiFi.

Goal

Prove that **the smart TV can access the internet.** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 4 - Machine Auto Shutdown

Knowledge Base

1. If a machine overheats, then the machine shuts down automatically.
2. The machine overheated.

Goal

Prove that **the machine shuts down automatically.** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 5 - Vehicle Registration Process

Knowledge Base

1. If a car passes the emissions test, then the car is certified roadworthy.
2. If a car is certified roadworthy, then the car can be registered.
3. The car passed the emissions test.

Goal

Prove that **the car can be registered.** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause (Box).
5. Explain each resolution step and write the final conclusion.

...ICS FOR CALCULATION

Numerical Problems

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	15
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	19
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	24
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	19
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretatio n with minor omissions	Limited or unclear interpretation	No interpretation	14

=95

1: Account Verification:

1. propositional logic

let

P = user uploads a valid ID

Q = Account is verified.

Knowledge Base:

1. $P \longrightarrow Q$

2. P

Goal: prove Q

2. Convert to CNF

1) $P \rightarrow Q \equiv \neg P \vee Q$

Fact: P

Negate goal: $\neg Q$

Clauses:- C_1: $\neg P \vee Q$

C_2: P

C_3: $\neg Q$

3. Resolution :-

From C_1 and C_2

$(\neg P \vee Q), P \Rightarrow Q$

Now resolve Q with $\neg Q$:-

$Q, \neg Q \Rightarrow \square$

4. Empty clause: $\square$

Hence the account is verified.

2. Premium Subscription Access :-

1. propositional logic :-

Let

P = Anitha pays subscription fee

Q = Anitha get premium Access

Knowledge Base :

1. $P \to Q$
2. P

Goal : Q

2. CNF :-

1) $P \rightarrow Q \equiv \neg P \vee Q$

Fact : P

Negated goal : $\neg Q$

Clauses :-

1) $C_1 : \neg P \vee Q$

2) $C_2 : P$

3) $C_3 : \neg Q$

3. Resolution

$(\neg P \vee Q), P \Rightarrow Q$

$Q, \neg Q \Rightarrow \square$

4. Empty clause : $\square$

Conclusion :- Anitha get premium Access.

3. Smart Device Connectivity:-

1. propositional logic

let

W = Device is connected to wifi

I = Device can access the internet.

Knowledge Base!

1) $W \rightarrow I$

2) W → Smart TV is connected to wifi

2. CNF :-

1) $W \rightarrow I \equiv \neg W \vee I$

Fact : W

Negated goal : $\neg I$

3. Resolution:-

$\{\neg W \vee I\}, W \Rightarrow I$

$I, \neg I \Rightarrow \square$

4. Empty clause:- $\square$

conclusion :- The Smart TV can access the internet.

4. Machine Auto shutdown:-

1. propositional logic

Let

O = Machine overheated

S = Machine shuts down automatically.

Knowledge Base:-

1. $O \rightarrow S$
2. O

Goal: S

2. CNF:-

1) $O \rightarrow S \equiv \neg O \vee S$

Fact: O

Negated goal: $\neg S$

Clauses:-

C_1: $\neg O \vee S$

C_2: O

C_3: $\neg S$

3. Resolution :

$(\neg O \vee S), O \Rightarrow S$

$S, \neg S \Rightarrow \square$

4. Empty clause : $\square$

Conclusion :- The machine shuts down auto-matically

5. Vehicle Registration Process :-

1. propositional logic

Let

E = Car passed the emissions Test

C = Car is certified roadworthy

R = Car can be registered.

Knowledge Base :-

1) $E \rightarrow C$

Goal : R

2) $C \rightarrow R$

3) E

2. <u>Convert to CNF</u>:-

1) $E \rightarrow C \equiv \neg E \vee C$

2) $C \rightarrow R \equiv \neg C \vee R$

Fact : E

Negated Goal : $\neg R$

Clauses:-

C_1 : $\neg E \vee C$

C_2 : $\neg C \vee R$

C_3 : E

C_4 : $\neg R$

3. Resolution:-

1) Resolve C_1 & C_3:-

$(\neg E \vee C), E \Rightarrow C$

2) Resolve the result with C_2:

$C, (\neg C \vee R) \Rightarrow R$

3) Resolve R with C_4:

$R, \neg R \Rightarrow \square$

4) Empty clause : $\square$

5. Final conclusion :-

Since the empty clause $\square$ is obtained, the negation of the goal is contradicted.

Therefore the case can be registered.